Edge-Guided and Multiscale Feature Optimization-Based Semantic Segmentation Network for Remote Sensing Images

Cheng Zhou, Bofeng Cai, Chengyi Xiong, *Senior Member, IEEE*, and Renfeng Liu

***Abstract*—Accurate semantic segmentation of remote sensing imagery is fundamental to geospatial analysis, yet the task remains daunting due to intrinsic complexities such as blurred object boundaries, severe occlusion, and high interclass similarity. To overcome these persistent bottlenecks, this study presents an engineering-integrated framework that strategically synthesizes advanced vision components to balance segmentation accuracy with model efficiency. Rather than relying on a single core innovation, we rationally combine a pyramid vision transformer (PVT) backbone with specialized edge and alignment modules to reconcile the tradeoff between global context and local detail. Specifically, the architecture features an edge-guided encoder that augments the PVT backbone with a learnable edge-guided feature module (EFM), explicitly targeting the recovery of boundary information often lost during downsampling. Complementing this, the decoder employs a feature-aligned pyramid network (FaPN) reinforced by adaptive spatial feature fusion (ASFF). This combination ensures precise cross-scale alignment and dynamic feature selection, effectively suppressing background noise while preserving fine-grained spatial structures. Experimental results on two large-scale high-resolution remote sensing datasets demonstrate that the proposed network offers superior segmentation performance while maintaining an efficient balance between computational complexity and model parameters.**

***Index Terms*—Boundary guidance, deep convolutional neural networks (CNNs), multiscale features, semantic segmentation.**

I. Introduction

REMOTE sensing image segmentation is a key component in the interpretation of geospatial information, and its accuracy directly affects the reliability of downstream tasks such as smart city management, land use classification, and environmental monitoring [1], [2]. Early techniques mainly relied on spectral features and machine learning algorithms, such as support vector machines [3] and random forests [4]. However, these traditional methods were limited by manually designed feature engineering and shallow model structures, making it difficult to capture complex nonlinear relationships or generalize across diverse scenes. This limitation rendered deep learning-based solutions an inevitable choice to break through existing bottlenecks.

The advent of deep learning, marked by the introduction of fully convolutional networks [5], brought a paradigm shift to the field. Early adaptations of convolutional neural network (CNN) architectures, such as U-Net [6] and DeepLabv3+ [7], significantly improved end-to-end learning capabilities. More recently, the integration of Transformer-based architectures has further pushed the boundaries of performance by modeling global dependencies [8]. Despite the success of these advanced models, they often underperform on high-resolution remote sensing data, which is characterized by extreme scale variations among objects (e.g., from meter-scale vehicles to kilometer-scale fields) and ambiguous regional boundaries. These factors complicate the segmentation process [9], where the variability in scale and blurred transitions continue to present persistent difficulties in achieving accurate and robust semantic segmentation.

The latest research shows that current remote sensing image segmentation methods face three core limitations that need to be overcome urgently: low interclass separability, high intraclass variation, and boundary adhesion issues, especially in complex scenes such as plastic greenhouse extraction [10]. The inherent hierarchical downsampling mechanism and local receptive field characteristics of CNNs further aggravate boundary attenuation through reduced spatial resolution and loss of contextual information [11]. In terms of scale adaptability, the extreme differences in the size of remote sensing targets are fundamentally inconsistent with the fixed receptive field architecture [12]. Although the traditional multiscale processing method introduces the feature pyramid network (FPN) to construct a hierarchical feature expression, its standard implementation scheme will cause boundary distortion and semantic inconsistency when multilevel features are fused due to the spatial misalignment of cross-scale feature maps in the upsampling stage [13]. Crucially, these challenges are interdependent: blurred boundaries often coexist with multiscale objects (e.g., small vehicles adjacent to large buildings), making isolated solutions insufficient. A model that only enhances edges may still fail to detect small targets if deep

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features dominate fusion; conversely, a model with strong multiscale fusion may smooth out critical boundaries if low-level details are not preserved or weighted adaptively. Therefore, our design is motivated by the need for a unified framework that simultaneously reinforces edge-aware representations to mitigate boundary blurring while enabling context-aware and adaptive fusion of multiscale features to maintain both fine structural details and semantic consistency. This dual-path optimization serves as the central rationale behind our edge-guided encoder and multiscale feature optimization decoder.

In addition, the inevitable information loss in the CNN architecture causes the high-level feature maps after the pooling operation to lose fine-grained spatial details [14], resulting in mediocre performance in refined tasks such as vegetation recognition and building facade contour extraction. To overcome the above bottlenecks, the Visual Transformer (ViT) establishes pixel-level global dependencies through a self-attention mechanism [15], completely breaking the local receptive field limitation of traditional CNNs and realizing cross-region context modeling in complex scenes with intertwined objects. In addition, the pyramid visual transformer (PVT) innovatively combines the Transformer's global perception advantage with a multiscale pyramid structure [16], effectively capturing cross-scale semantic information from micro to macro in remote sensing images through hierarchical feature interaction. To address the problem of blurred edge segmentation, the edge guidance mechanism strengthens the feature response of the contour region through a learnable boundary weight matrix [17], significantly improving the segmentation accuracy of irregular object boundaries. The feature alignment pyramid network (FaPN) designs a cross-scale deformable convolution alignment module [18], fundamentally solving the fusion artifact problem caused by the spatial misalignment of traditional FPN feature maps and ensuring the geometric consistency of multilevel features. The adaptive spatial feature fusion (ASFF) mechanism realizes context-aware fusion of multiscale features through dynamic weight learning [19], optimizing edge continuity while improving the sensitivity of small target detection. These innovative architectures jointly promote the evolution of remote sensing image segmentation technology toward higher accuracy and robustness through collaborative optimization of global modeling, multiscale adaptation, and feature preservation. Despite recent advancements in remote sensing image segmentation, several critical challenges remain.

1) Object boundaries in remote sensing images are often blurred and irregular due to limited resolution, sensor noise, or complex scene structures, hindering accurate contour extraction.
2) Partial or full occlusion by other objects or terrain elements reduces the model's ability to identify and separate target regions.
3) Objects from different categories may share similar appearances, leading to misclassification and decreased segmentation accuracy.

In response, this study presents an engineering-integrated framework tailored to reconcile these conflicts. Rather than relying on a single theoretical innovation, we focus on the strategic synthesis of complementary advanced modules. The architecture anchors a PVT backbone with a learnable edge-guided feature module (EFM) to explicitly recover boundary cues lost during downsampling. Complementing this, the decoder employs a feature-aligned pyramid network (FaPN) reinforced by ASFF. This combination ensures that deep semantic features are rigorously aligned with shallow spatial details, enabling the model to distinguish fine-grained targets even in complex scenes.

We emphasize that the primary contribution of this work is the proposal of an engineering-integrated remote sensing segmentation framework. Rather than introducing fundamentally new theoretical operators, the novelty lies in the strategic adaptation and synergistic integration of complementary components—PVT backbone, learnable EFM, feature alignment decoder (FaPN), and ASFF. This design is explicitly motivated by the practical engineering challenge of reconciling the scale-boundary tradeoff in high-resolution remote sensing imagery. The scientific value stems from combining these design choices to jointly improve boundary fidelity and multiscale fusion efficiency. To sum up, the major contributions of this work are summarized as follows.

1) We propose an engineering-integrated network that validates how the rational combination of edge-guidance and multiscale alignment can achieve comprehensive performance gains, effectively balancing segmentation accuracy with inference efficiency.
2) An edge-guided encoder that integrates a PVT backbone with the proposed EFM to enhance edge feature representation and alleviate boundary blur.
3) A multiscale feature optimization decoder is introduced for improved spatial detail reconstruction and better integration between deep semantic and shallow spatial features. It leverages FaPN and an improved ASFF module to accomplish more precise multiscale feature alignment and fusion.

II. Related Work

A. Transformer-Based Architectures for Remote Sensing

Hierarchical vision transformers (ViTs) have largely superseded traditional CNNs in remote sensing by explicitly modeling the long-range dependencies essential for interpreting complex landscape patterns [20]. Notably, the pyramid vision transformer (PVT) facilitates efficient multiscale feature extraction [21], with domain-specific adaptations proving effective in tasks ranging from vegetation mapping [22] to fine-grained context modeling [23]. While updated iterations like PVTv2 [16] offer improved computational efficiency, standard transformer architectures inherently struggle to preserve local high-frequency details [24]. This limitation often leads to failures in resolving the heterogeneous object boundaries common in satellite imagery [25]. Although recent hybrid attempts such as SwinFAN [26] and Cascaded Transformer [27] aim to bridge this gap by integrating guided attention [28], they frequently fall short in achieving precise edge segmentation [29]. This persistent deficiency motivates the integration of

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dedicated boundary-preservation mechanisms into the feature learning pipeline.

B. Edge-Guided Feature Learning

Preserving boundary sharpness in complex urban environments remains a substantial hurdle for conventional encoder-decoder architectures, which frequently struggle to delineate irregular structural forms [30]. While early methodologies incorporated explicit boundary detection networks [31] or boundary-aware transformers [32] to enhance precision, these approaches often necessitate separate processing stages or complex preprocessing. Consequently, recent research has shifted toward embedding boundary awareness directly into the learning framework. Strategies such as dual-domain transformers [33] and dual-task learning [34] have been developed to jointly optimize segmentation and edge objectives. Furthermore, advanced designs like the boundary-enhanced dual-stream network (BEDSN) address boundary ambiguity in high-resolution imagery [35], while dynamic snake convolutions have been integrated into SegFormer to better capture slender geometric features [36]. Despite these advances, most existing methods still treat edge information primarily as an auxiliary supervision signal rather than utilizing it as an intrinsic component of feature construction.

Another critical challenge stems from the progressive degradation of edge information during successive downsampling. Although reinforcement strategies attempt to reintroduce spatial details into deeper network stages [37], maintaining fine granularity remains difficult. This issue is particularly pronounced in transformer-based models, where global self-attention mechanisms tend to inadvertently suppress local spatial cues [38]. While recent frameworks like the Boundary-Aware Mask2Former seek to mitigate this via improved feature bridging [39], effectively balancing global context modeling with pixel-level boundary precision, without incurring excessive computational overhead, remains an open research question.

Compared to existing boundary-aware methods, our approach is distinguished by three key aspects. First, it learns multiscale edge priors through an edge feature module (EFM) that directly extracts edge representations from intermediate backbone features without relying on external edge maps or manually designed detectors. Second, it performs alignment-prioritized fusion, where cross-scale features are explicitly aligned before fusion to mitigate spatial misalignment and reduce boundary artifacts. Third, it adopts space-adaptive fusion, which assigns pixel-wise weights to adaptively emphasize the most informative scale at each spatial position, thereby improving the preservation of small objects and boundary details.

C. Feature Alignment and Pyramid Networks

Feature misalignment during multiscale fusion represents a significant obstacle in segmentation models. Although the FPN is widely employed for integrating multiscale features [40], standard implementations frequently suffer from imprecise spatial correspondence across different resolutions [41]. This limitation leads to boundary degradation, which is particularly detrimental in remote sensing contexts characterized by fine-grained and heterogeneous land cover elements [42]. To mitigate these issues, recent research has augmented pyramid structures with explicit alignment and enhancement strategies. For example, the high-resolution feature pyramid attention network (HRFPANet) utilizes attention mechanisms to improve classification in complex zones [43], while the multistage feature enhancement pyramid network (FEPN) refines features to optimize object detection [44]. Furthermore, architectures such as the location refined FPN (LR-FPN) have been developed to ensure more accurate scale-wise correspondence during fusion [45]. Despite these innovations, effectively adapting alignment strategies to the substantial scale variability and intricate spatial distributions of remote sensing imagery remains difficult [46].

Current approaches typically face a tradeoff between performance and efficiency. Attention-guided fusion modules [47] can improve spatial consistency but often incur high computational costs that limit scalability [48]. Conversely, lightweight alignment mechanisms designed to reduce overhead [49] frequently struggle to segment small or fragmented objects accurately. Given the necessity of maintaining precise feature alignment for discontinuous land cover types [50], our work introduces a FaPN. This approach aims to ensure geometric consistency without imposing the heavy computational burden associated with extensive attention mechanisms.

D. Adaptive Feature Fusion

Accurate segmentation across diverse object scales in remote sensing imagery relies heavily on the effective integration of multiscale features. Traditional static fusion strategies have been shown to be insufficient for capturing the spatial and contextual variability inherent in overhead imagery, often leading to suboptimal segmentation outcomes [51]. ASFF, originally developed for object detection tasks, has demonstrated considerable potential when applied to semantic segmentation, offering improved flexibility in handling spatial heterogeneity [52]. Adaptations of ASFF for land cover classification have resulted in notable enhancements in boundary precision and the detection of small-scale objects [53].

Despite these improvements, managing the extreme scale variation typical of remote sensing data remains difficult [54]. Consequently, recent studies have prioritized dynamic, context-aware strategies. For example, the Adaptive Frequency Enhancement Network (AFENet) modulates frequency information to improve cross-scale interaction [55], while PPCNet employs panoramic fusion to capture global contexts [56]. Additionally, AdaptiveFusionNet utilizes attention mechanisms to selectively emphasize informative regions [57]. These approaches represent significant progress by dynamically modulating feature contributions according to semantic content [58] and object-specific characteristics [59].

However, a limitation persists as current adaptive fusion methods rarely prioritize boundary-relevant features explicitly. Most existing mechanisms focus on regional consistency, which often compromises edge fidelity in the final output. This

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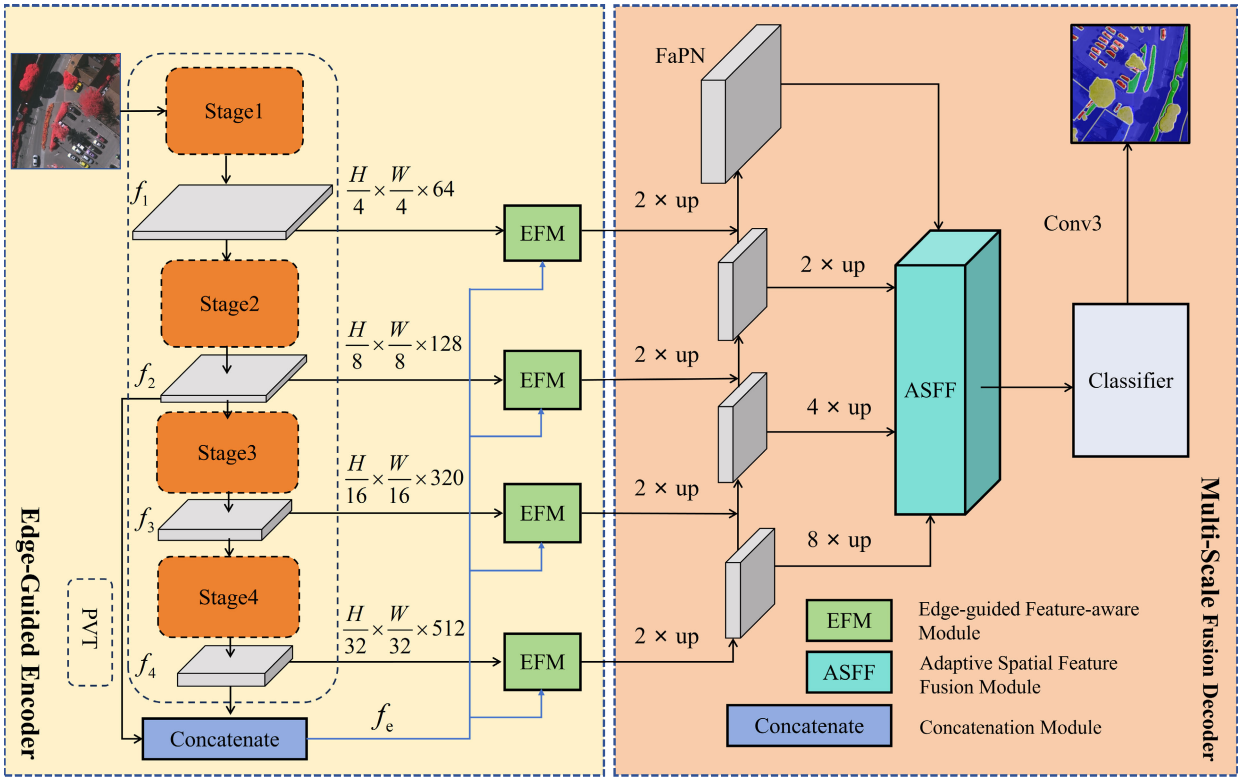


Fig. 1. Overall architecture of the edge-guided multiscale feature optimized remote sensing image semantic segmentation network.

reduction in precision is particularly detrimental to applications requiring exact delineation, such as urban infrastructure mapping [60]. To overcome this, our proposed method integrates a boundary-aware mechanism into the adaptive fusion process, ensuring the preservation of edge details alongside deep semantic features.

III. Methodology

A. Overall Network Architecture

As shown in Fig. 1, the edge-guided multiscale feature optimization network comprises two main components: an edge-guided encoder and a multiscale feature optimization decoder. First, the edge-guided encoder utilizes PVT for multiscale feature extraction, which efficiently captures multiscale features. An integrated EFM is incorporated to enhance edge feature representation, effectively reducing boundary blurring and improving the delineation of object edges.. Subsequently, the multiscale feature optimization decoder is redesigned based on the FaPN structure to improve boundary segmentation performance. Additionally, an ASFF module is introduced within the decoder to dynamically optimize feature aggregation across scales. This module further enhances boundary delineation by refining boundary prediction, significantly improving the network's ability to capture intricate details and contours in the imagery.

The detailed process flow is as follows: Given a high-resolution remote sensing image, it is first fed into the PVT module. The input image initially passes through a stage where it is partitioned into small patches via a Patch Partition module. Each patch undergoes linear embedding to generate initial features. The PVT employs a self-residual attention (SRA) mechanism that integrates global context with local attention, progressively extracting both shallow detail features and deep semantic information across layers. Through the PVT backbone, multiscale feature maps f_1, f_2, f_3 and f_4 are obtained, forming a pyramid of features. Subsequently, these shallow and deep features are separately input into the EFM to enhance boundary feature expression, effectively alleviating the common problem of boundary blurring in segmentation. Next, the extracted multiscale features are fed into the FaPN decoder. FaPN employs layer-wise upsampling with explicit feature alignment across scales, facilitating the rich fusion of high-level semantic information with low-level spatial details, thereby substantially improving segmentation accuracy. The fused features are further processed through an ASFF module, which adaptively combines spatial information across scales, enhancing the network's ability to capture small targets and complex boundary structures. This ensures high-precision segmentation results. Finally, the optimized features are passed through a convolutional layer (Conv3) and a classifier to generate the final semantic segmentation output.

B. Design of the Edge-Guided Encoder Module

Due to its outstanding performance in global context modeling, ViT models have demonstrated significant effectiveness in remote sensing semantic segmentation tasks. However, the high resolution and complex scenes typical of urban remote sensing images lead to substantial computational costs in Transformer-based models. The increased complexity of training and inference, along with the limitations of using a

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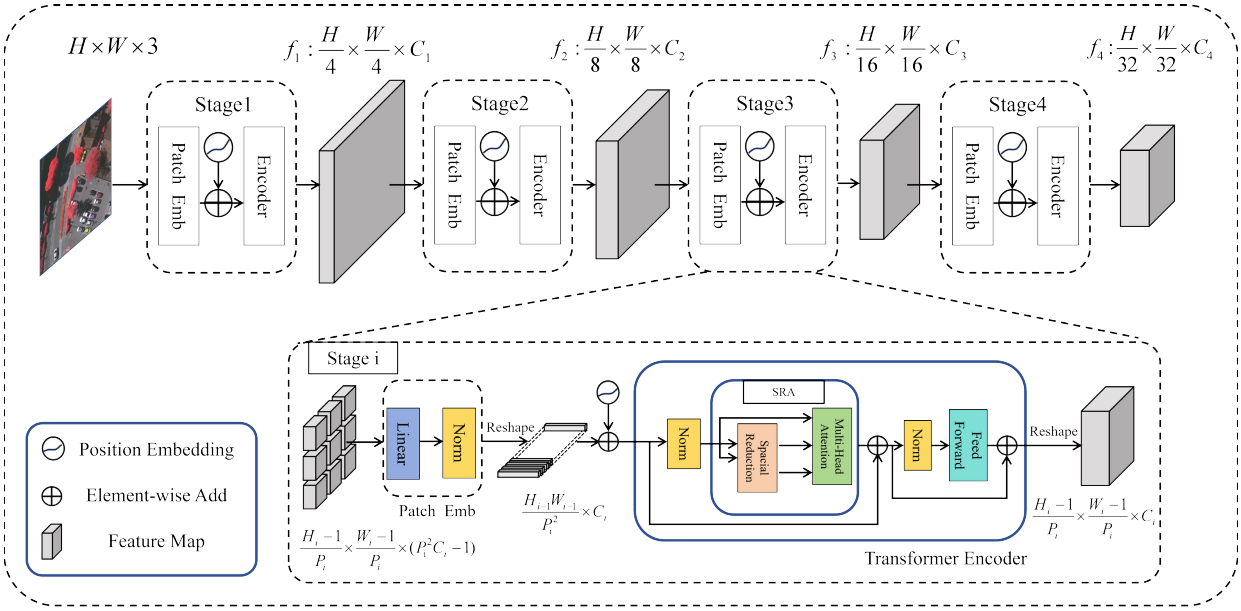


Fig. 2. Overall architecture of the PVT-based encoder.

single global self-attention mechanism, hinders performance in practical applications, where the efficiency loss outweighs the benefits of marginal accuracy improvements.

To overcome these challenges, this work employs a highly efficient PVT as the backbone network, which progressively extracts multiscale features through a pyramid structure. By integrating a SRA mechanism, PVT effectively reduces computational complexity while enhancing the representation capacity for targets at different scales. Its multiscale feature extraction capabilities make it particularly suitable for high-resolution, complex-scene image segmentation, offering a more practical and efficient alternative to traditional Transformers. To further enhance the model's ability to preserve edge details in high-resolution remote sensing images, an EFM is integrated into the encoder, forming a novel edge-guided encoder. This encoder encompasses two key design aspects: first, employing PVT as the backbone to capture and extract rich multiscale features while simultaneously reducing computational complexity, thereby improving efficiency and resource utilization. Second, adding an EFM after each feature extraction stage to reinforce edge feature representation, which effectively mitigates boundary blurring caused by complex backgrounds and noise interference typical in remote sensing imagery. The overall architecture of the PVT-based encoder is illustrated in Fig. 2.

The overall architecture of PVT consists of four similar stages, each composed of multiple identical Transformer encoder modules. Each encoder module comprises two primary components: the multihead self-attention (MHSA) layer and the feedforward multilayer perceptron (FMLP), forming the core mechanisms for feature extraction and representation. The computation of MHSA is described as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{1}$$

where Q, K, and V denote the query, key, and value matrices, respectively.

The PVT adopts a hierarchical feature pyramid construction strategy by progressively reducing the input resolution through staged Patch Embedding operations. The specific implementation process is as follows: In the initial stage, a remote sensing image of size $H \times W \times 3$ is divided into $(H/4) \times (W/4)$ image patches, each with a spatial size of $4 \times 4 \times 3$. These patches are flattened and then passed through a linear transformation to generate patch embedding features of size $(H/4) \times (W/4) \times C_1$. After fusion with positional encodings, these features are input into a Transformer encoder with $H \times W \times 3$ layers, producing the feature map f_1 with dimensions $(H/4) \times (W/4) \times C_1$. Subsequent stages utilize a cascade approach, where the feature map from the previous stage serves as input for the next, following the same process to generate features f_2, f_3, and f_4. The spatial downsampling rates for these feature maps are 8×, 16×, 32×, and 64×, respectively. This multiscale feature extraction architecture makes PVT an ideal backbone network for remote sensing image segmentation tasks.

To process high-resolution remote sensing features more efficiently, PVT introduces a linear SRA mechanism by modifying the multihead attention (MHA) modules in ViT. SRA reduces the spatial dimensions of the Q, K, and V vectors through downsampling, significantly lowering computational complexity and memory consumption. The first stage of SRA can be formulated as

$$\text{SRA}(Q, K, V) = \text{Concat}\left(\text{head}_0, \text{head}_1, \ldots, \text{head}_{N_i}\right) W^o \tag{2}$$

$$\text{head}_j = \text{Attention}\left(QW_j^Q, \text{SR}(K) W_j^K, \text{SR}(V) W_j^V\right) \tag{3}$$

$$\text{SR}(x) = \text{Norm}\left(\text{Reshape}(x, R_i) W^S\right) \tag{4}$$

where W is the linear projection matrix, N is the number of attention heads, i indicates the stage, SR is the spatial reduction strategy, R is the reduction ratio, and the computational complexity of the Attention mechanism decreases by a factor of R^2 relative to the original ViT.

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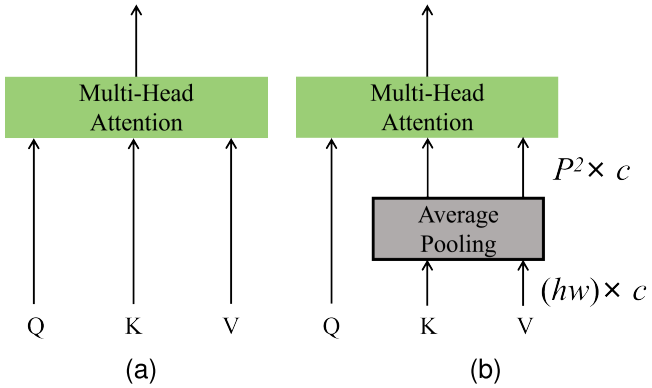


Fig. 3. Structure of (a) MHA and (b) linear spatial reduction attention.

In this work, we adopt the upgraded version of PVT, PVTv2 [16], which is characterized by the introduction of a linear SRA mechanism. SRA reduces the computational and memory costs by performing spatial downsampling on the K and V features. Compared to the conventional MHA, this approach significantly improves efficiency. Specifically, the attention complexity is decreased from quadratic to linear, while still maintaining the ability to capture global contextual information. This optimization enables PVTv2 to better meet the demands of high-resolution remote sensing image processing. The operational differences between SRA and traditional MHA are illustrated in Fig. 3, highlighting notable distinctions in computational process and resource consumption.

Compared to earlier versions, PVTv2 introduces significant innovations in the SRA mechanism. Specifically, the linear SRA module replaces traditional convolution operations with an average pooling strategy. It uniformly downsamples the feature space dimensions from $H \times W$ to a fixed size P^2 prior to attention computation. This modification not only streamlines the operation process but also ensures that the computational complexity and memory consumption of the linear SRA remain comparable to those of convolutional layers, further enhancing the model's efficiency.

In the feature fusion stage, the initial features extracted by PVTv2 are fed into the EFM. The proposed EFM is a fully learnable, end-to-end module that operates without predefined edge detectors. This design enables the network to inherently focus on boundary regions without requiring manual intervention or external edge supervision. The processing steps are summarized as follows.

1) *Multiscale Stitching:* Feature maps (f_1, f_2, f_3, and f_4) extracted from PVTv2 serve as connecting channels to aggregate boundary cues across multiple scales.
2) *Feature Enhancement:* Residual connections and 3×3 convolutions are applied to refine the extracted features, while a local attention-based channel weighting mechanism further emphasizes critical information through adaptive channel modulation.
3) *Boundary Feature Generation:* As illustrated in Fig. 4, the input features f_i (where $i = 1, 2, 3, 4$) are concatenated to form the boundary feature f_e. When $i \geq 3$, f_e is downsampled (denoted as D) to match the spatial resolution of f_i, after which it is multiplied with each f_i individually.

This result is further combined through additional residual connections and element addition, and then subjected to a 3×3 convolution to obtain the initial fused feature map. The operation summary is as follows:

$$f_i^e = \begin{cases} F_{\text{conv}}\left(\left(f_i \otimes f_e\right) \oplus f_i\right), & i = 1, 2 \\ F_{\text{conv}}\left(\left(f_i \otimes D\left(f_e\right)\right) \oplus f_i\right), & i = 3, 4 \end{cases} \tag{5}$$

where D indicates the downsampling operation used to align the resolution of the boundary feature f_e with the lower-resolution features f_3 and f_4. F_{conv} represents a 3×3 convolution operation, $\otimes$ indicates element-wise multiplication, and $\oplus$ denotes element-wise addition.

To enhance the feature representation capability, a local attention mechanism is introduced to capture key feature channels. First, a channel-wise global average pooling (GAP) is applied to aggregate the convolutional features. Next, a 1-D convolution combined with a Sigmoid activation function is used to generate corresponding channel weights. Unlike fully connected layers, which can model dependencies among all channels but involve higher computational complexity, this approach employs a local cross-channel interaction strategy that considers only the relationships between each channel and its k neighboring channels for localized learning. Finally, the generated channel attention is multiplied element-wise with the original input feature f_i^e, and a 1×1 convolution is applied to compress the channel dimension, producing the final output f_i^a

$$f_i^a = F_{\text{conv1}}\left(\sigma\left(F_{1-D}^k\left(\text{GAP}\left(f_i^e\right)\right)\right)\right) \otimes f_i^e \tag{6}$$

where F_{conv1} denotes the 1-D convolution with kernel size k, and σ is the Sigmoid function. And the kernel size $k = |(1 + \log_2 C)/2|_{\text{odd}}$ can be adaptively set to an odd number, where $|*|_{\text{odd}}$ denotes the nearest odd integer. The kernel size is proportional to the number of channels C of f_i^e. This attention strategy effectively emphasizes important channels while suppressing noise, thereby enhancing the semantic representation.

C. Design of the Multiscale Fusion Decoder Module

In remote sensing image processing, the diverse sizes of land cover objects and the prevalence of small targets often lead to significant loss of critical details when relying solely on deep-layer network features for segmentation. These details are especially vital for accurately segmenting small objects and edge regions. To address this, we propose a multiscale feature fusion decoder. This decoder leverages the architectural advantages of the FaPN to effectively fuse detailed information from various hierarchical levels of the network with high-level semantic features. Additionally, it incorporates an ASFF module, which employs an adaptive attention mechanism to perform weighted fusion of these features, better preserving important features, reducing noise, and extracting richer contextual information. This approach enhances the perception of small targets and improves boundary segmentation accuracy.

Specifically, features A1, A2, A3, and A4 are extracted from the encoder at multiple scales. These features are aligned

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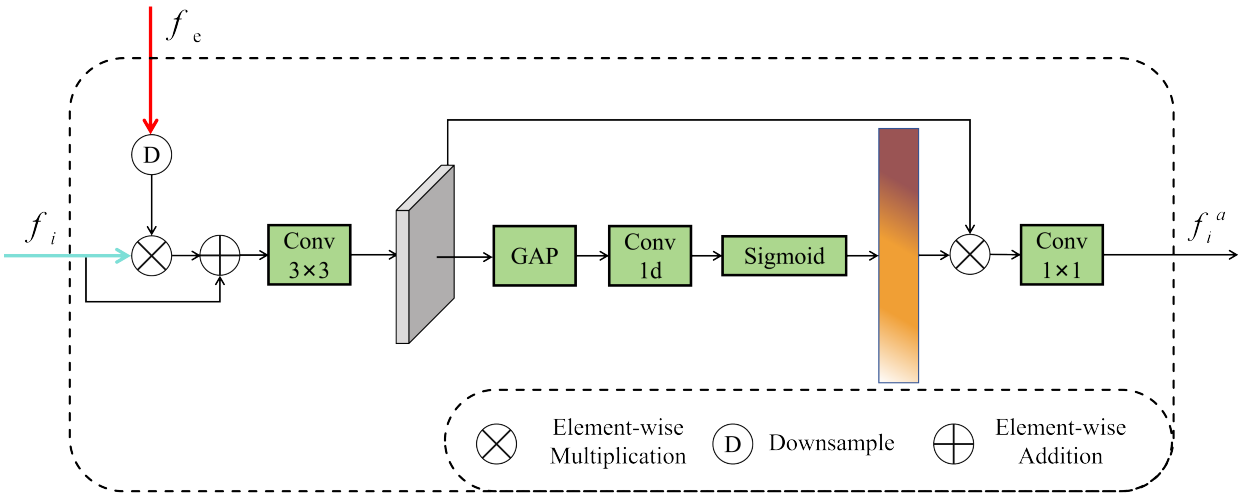


Fig. 4. Structure of the EFM.

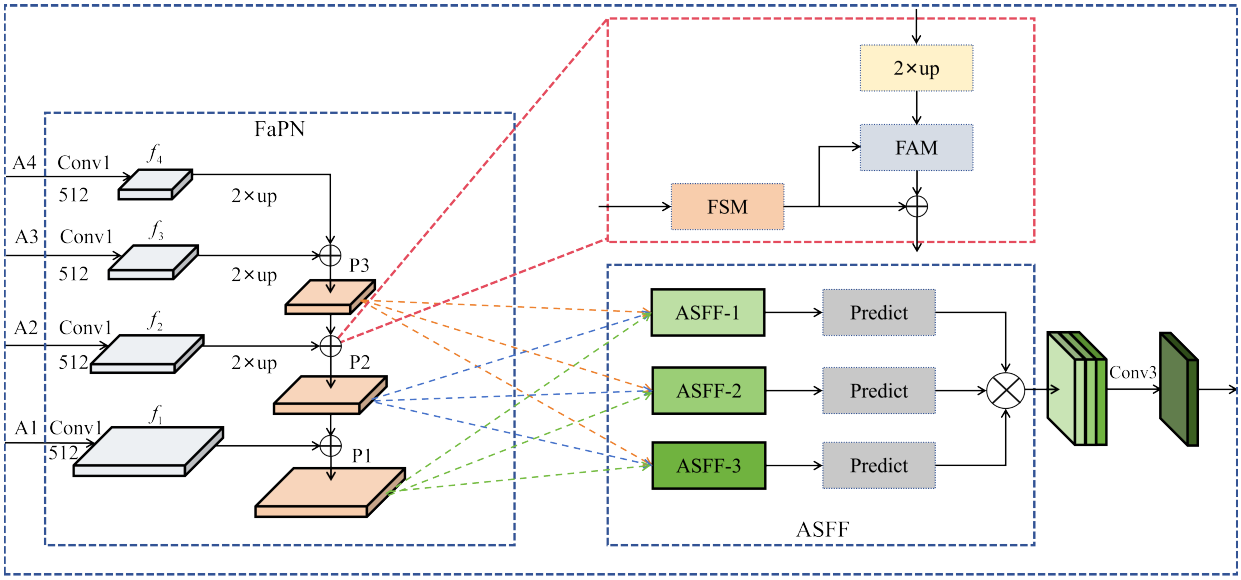


Fig. 5. Structure of the multiscale fusion decoder module.

through 1 × 1 convolutions to unify their spatial resolution and channel dimensions, generating features P1, P2, and P3. The feature selection module (FSM) dynamically selects key information across scales using attention mechanisms. Given input features *Pi* (i = 1,2,3), FSM computes channel attention weights, which are used to reweight the features accordingly. These reweighted features are then passed to the Fusion and Attention Module (FAM), which further fuses features across scales, aggregating both global and local information. After the features are aligned to the same resolution, a pixel-wise addition fusion is performed with the output from FSM. The fused features are then processed through the ASFF module for adaptive weighting, further refining the features to generate the final segmentation output. The overall multiscale fusion decoder module is illustrated in Fig. 5.

In the FPN, directly adding the upsampled features to those extracted from the backbone may result in spatial misalignments of the contextual feature maps. This misalignment can lead to segmentation and classification errors, particularly along object boundaries. In contrast to standard FPN implementations where feature selection and spatial alignment are often intertwined, we explicitly decouple these operations within the FaPN architecture. This design adheres to a strict "select-then-align" protocol, which is essential for handling the complex backgrounds typical of remote sensing imagery. Functioning as a semantic prefilter, the FSM first suppresses redundant channel responses and background noise. This step is critical to ensure that the input to the subsequent stage consists solely of discriminative signals. Once purified, these features are processed by the feature aggregation module (FAM), which focuses exclusively on spatial alignment and multiscale aggregation. By isolating the selection process from spatial transformation, we effectively halt the propagation of noise across scales, preventing the network from aligning and reinforcing irrelevant background clutter.

The FSM employs an attention mechanism to dynamically select critical information across different scales. Given input

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features P_i, FSM first computes the channel attention weights as follows:

$$W_i = \sigma\left(\mathrm{FC}\left(\mathrm{GAP}\left(P_i\right)\right)\right) \tag{7}$$

where GAP denotes the global average pooling operation, FC is a fully connected layer, and σ represents the Sigmoid activation function. The features are then reweighted via element-wise multiplication, yielding

$$F_{\mathrm{FSM},i} = P_i \odot w_i \tag{8}$$

where $\odot$ indicates element-wise multiplication.

The FAM fuses features across multiple scales, integrating both global and local information. The features outputted by the FSM, denoted as F_{FSM}, are first adjusted to a uniform resolution via upsampling or downsampling, represented by the resize operation. These features are then fused through pixel-wise addition

$$F_{\mathrm{FAM}} = \sum_{i=1}^{N} \mathrm{Resize}\left(F_{\mathrm{FSM},i}\right) \tag{9}$$

where Resize indicates the operation to unify the feature map resolutions. The combination of FSM and FAM enables the FaPN to efficiently extract salient features and aggregate multiscale semantic information. Ultimately, each level of FaPN's output is fed into the ASFF module for weighted fusion, producing the final segmentation results.

The ASFF module introduces an adaptive structure for feature fusion, designed to integrate features from multiple levels using element-wise or cascaded multilevel fusion methods. The essence of ASFF lies in its step-by-step implementation of feature fusion and the explicit mechanism for preserving small targets. For each output level, three aligned inputs $X(1)$, $X(2)$, and $X(3)$ (adjusted to the same spatial resolution for P1–P3) are fed into the fusion branch. A set of 1×1 convolutions followed by softmax normalization generates three spatial weight maps α, β and γ. These maps adaptively control the contribution of each feature source at every spatial location, ensuring that information from higher-resolution maps is not overwhelmed by coarser features. In this way, fine-grained details critical to small target detection are retained during fusion. The fused representation, termed ASFF-3, is then computed as a weighted sum of the aligned inputs. This process is formalized as

$$y_{ij}^{l} = \alpha_{ij}^{l} \cdot X_{ij}^{1\to l} + \beta_{ij}^{l} \cdot X_{ij}^{2\to l} + \gamma_{ij}^{l} \cdot X_{ij}^{3\to l} \tag{10}$$

where y_{ij}^{l} represents the new feature map obtained by ASFF. The parameters α_{ij}^{l}, β_{ij}^{l} and γ_{ij}^{l} denote the learned weight coefficients corresponding to the respective feature maps. The weights α, β and γ are obtained by passing the downsampled feature maps from levels 1 to 3 through 1×1 convolutions. These parameters are concatenated, and a softmax function normalizes them to ensure each weight falls within the range [0, 1] and their sum equals 1, as expressed in

$$\alpha_{ij}^{l} = \frac{e^{\lambda_{\alpha_{ij}}^{l}}}{e^{\lambda_{\alpha_{ij}}^{l}} + e^{\lambda_{\beta_{ij}}^{l}} + e^{\lambda_{\gamma_{ij}}^{l}}}. \tag{11}$$

Finally, through convolution operations, the features from different stages and modules are integrated to produce the final prediction result. Different from FPN, which integrates features through fixed lateral addition across all spatial positions, ASFF applies pixel-by-pixel adaptive weighting. This design allows boundary clarity and small target signals to be better preserved, while keeping the computational cost moderate, since the weight maps are generated by lightweight 1×1 convolutions followed by a softmax operation.

IV. Experimental Results

A. Dataset and Experimental Setting

1) Dataset: This study utilizes the ISPRS Potsdam dataset [61]. and the LoveDA dataset [62] to evaluate the accuracy and feasibility of the proposed algorithm.

Potsdam Dataset: [61]: This dataset comprises 38 high-resolution aerial images, each with a spatial size of 6000×6000 pixels. Every pixel has been semantically annotated with six types of land-cover categories: impervious surfaces (imp surf), buildings, low vegetation (low veg), trees, cars, and clutter/background (clutter). In the experiments, the original 38 images were cropped into smaller patches of 512×512 pixels, resulting in a total of 7488 sample images. The data were divided into 3456 training images, 2016 testing images, and 2016 validation images.

LoveDA Dataset: [62]: This is a large-scale, high-resolution optical remote sensing dataset designed explicitly for semantic segmentation tasks. It contains 5987 aerial images sourced from the cities of Nanjing, Changzhou, and Wuhan, covering both urban and rural scenes. The dataset features diverse remote sensing scenarios and a wide variety of small targets, providing a suitable validation platform for the proposed algorithm. Among these images, 2713 depict urban scenes, and 3274 depict rural areas. The dataset annotates seven categories: building, road, water, barren, forest, agriculture, and background. The training set includes 2522 images with corresponding labels for model training and parameter tuning; the validation set contains 1669 images to evaluate model performance during training; while the test set comprises 1796 images.

In our experiments, we employed a combination of augmentation strategies, including flipping, rotation, and cropping to augment the Potsdam and LoveDA datasets, ultimately enhancing the model's generalization capability and segmentation performance.

2) Evaluation Metrics: In this study, the performance of the segmentation models is evaluated using the mean Intersection over Union (mIoU) and Intersection over Union (IoU) metrics, which serve as the primary indicators of accuracy and efficiency. The calculation formulas are as follows:

$$\mathrm{IoU} = \frac{P_{ii}}{\sum_{j=0}^{k} P_{ij} + \sum_{j=0}^{k} P_{ji} - P_{ii}} \tag{12}$$

$$\mathrm{mIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{P_{ii}}{\sum_{j=0}^{k} P_{ij} + \sum_{j=0}^{k} P_{ji} - P_{ii}} \tag{13}$$

where $\sum P_{ij}$ represents the total number of pixels that are truly of class i but incorrectly predicted as class j, and $\sum P_{ji}$ represents the total number of pixels that are truly of class j but incorrectly predicted as class i. The IoU for each class i is

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TABLE I
QUANTITATIVE COMPARISON RESULTS ON THE POTSDAM DATASET

Backbone	Method	IoU per category (%)						mIoU(%)	aAcc(%)
		Imp. surf.	Building	Low veg.	Tree	Car	Clutter		
CNN-base	FCN [5]	86.49	93.24	75.60	78.59	91.59	35.78	76.88	90.18
	DeepLab V3+ [7]	87.21	**93.69**	76.65	79.18	92.07	41.16	78.33	90.71
	Pspnet [63]	86.71	92.75	76.00	78.63	90.89	43.36	78.06	90.60
	DDRnet [64]	85.75	91.88	75.76	78.00	90.02	41.02	77.04	89.69
Transformer-base	Ster [19]	86.69	92.99	76.59	79.61	91.25	41.70	78.14	90.53
	Segmenter [65]	86.97	92.58	77.27	78.69	90.09	44.58	78.36	90.67
	Segformer [66]	87.07	93.58	76.99	79.54	92.30	42.05	78.59	90.78
	SAN [67]	85.54	91.58	75.68	76.57	87.87	42.64	76.65	90.06
	Ours	**87.97**	93.42	**77.35**	**79.94**	**92.48**	**46.70**	**79.64**	**91.39**

TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE LOVEDA DATASET

Backbone	Method	IoU per category (%)							mIoU%	aAcc%
		Background	Building	Road	Water	Barren	Forest	Agriculture		
CNN-base	FCN [5]	51.97	60.88	53.97	65.92	24.28	41.39	46.53	49.28	67.45
	DeepLab V3+ [7]	**54.17**	**63.8**	54.77	63.7	29.85	**42.33**	48.34	50.99	69.96
	Pspnet [63]	53.65	60.84	52.2	66.08	24.35	39.18	55.71	50.29	69.49
	DDRnet [64]	52.2	57.28	52.76	65.12	24.81	39.11	57.96	49.89	68.54
Transformer-base	Ster [19]	47.38	55.01	51.57	68.31	29.58	37.62	59.08	49.79	68.56
	Segmenter [65]	52.37	56.18	51.92	67.37	30.01	41.55	53.97	50.48	69.26
	Segformer [66]	51.77	56.17	52.38	71.19	31.76	40.29	58.64	51.74	70.4
	SAN [67]	51.38	62.7	50.81	64.23	30.13	37.56	54.48	50.2	68.71
	Ours	53.83	58.92	**54.94**	**71.22**	**31.84**	40.51	**59.38**	**52.95**	**71.49**

computed as the ratio of the intersection (correctly predicted pixels) to the union of the true and predicted pixels for class i; averaging these over all classes yields the mean IoU, providing a balanced measure of overall model performance.

Beyond these metrics, we introduce the Average Pixel Accuracy (aAcc) to more comprehensively evaluate the model's accuracy and efficiency. Their calculations are expressed as

$$\text{aAcc} = \frac{\sum_{i=0}^{k} P_{ii}}{\sum_{i=0}^{k} \sum_{j=0}^{k} P_{ij}} \tag{14}$$

where P_{ii} is the count of pixels accurately predicted as class i, and P_{ij} is the number of pixels with true label i predicted as class j. These metrics collectively enable a thorough evaluation of both the accuracy and efficiency of the models, providing critical insights for performance comparison and optimization.

3) Experimental Setting: The experimental implementation was carried out within the MMSegmentation framework, utilizing PyTorch version 1.11.0, mmcv-full 1.7.1, and Python 3.8. The computing environment was configured with Windows 10 as the operating system, CUDA 10.2 for GPU acceleration, and a GeForce GTX 1080 Ti GPU. The CPU used was an Intel i5-10400, and the deep learning training employed PyTorch 1.10.1 as the framework. For the training process, the stochastic gradient descent (SGD) optimizer was used to update network weights, with a momentum parameter of 0.9 and a weight decay of 1e-4. The initial learning rate was set to 0.0001. Training was conducted over 80 000 iterations, with each iteration corresponding to one batch of data.

During both training and testing phases, multiple data augmentation techniques were employed to improve model robustness and generalization. These included random horizontal flipping, random transposing, and random scaling, with the scaling factors sampled within the range of 0.5 to 2.0. These augmentations aimed to simulate diverse real-world conditions, thereby enhancing the model's ability to perform well across varied scenarios.

B. Quantitative Comparison of Diverse Methods

In this study, multiple experiments were conducted on the Potsdam and LoveDA remote sensing datasets. During training, the model weights corresponding to the optimal validation performance were preserved. The proposed method was compared with several mainstream algorithms on both datasets, including CNN-based approaches such as FCN [5], DeepLab v3 + [7], PSPNet [63], and DDRNet [64], as well as Transformer-based methods including Ster [19], Segmenter [65], SegFormer [66], and SAN [67]. The experimental results on the Potsdam dataset are summarized in Table I, while those on the LoveDA dataset are presented in Table II. In both tables, the highest performance metrics are highlighted in bold for clarity.

1) Quantitative Comparison of Results on the Potsdam Dataset: According to the data in Table I, the proposed model outperforms other models across several evaluation metrics.. Our approach achieves an mIoU and aAcc that surpass the best-performing CNN-based method, Deeplabv3+, by 2.66% and 1.53%, respectively. It also outperforms the best

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Transformer-based method, Segformer, by 1.29% in mIoU and 1.04% in aAcc.

Given the translational invariance inherent in convolutional operations, CNN-based methods have a notable advantage when recognizing large, regular objects like buildings, with IoU in the building category slightly higher than our method. However, when dealing with small-scale and irregular objects, this method exhibits superior performance. To verify this, we categorized targets by scale. Statistical analysis shows that for small targets, specifically Car and Clutter, our method achieves an average IoU improvement of 1.7% and 5.1%, respectively, compared to the average performance of other models in this experiment. In contrast, the improvements for medium-scale targets including Low Vegetation and Tree are 1.03% and 1.33%, respectively. Similarly, for large-scale targets such as Impervious Surfaces and Buildings, the improvements are 1.42% and 0.63%, respectively. Therefore, the overall segmentation effect on small-scale objects is more significant compared to other methods.

This superior performance can be attributed to several key factors. The PVTv2 backbone provides robust multiscale and global–local representations, effectively reducing semantic confusion among heterogeneous landscape elements compared to conventional CNN backbones. Meanwhile, the EFM enhances local contrast along object contours by explicitly extracting and reinjecting boundary-aware features at multiple encoder stages. This effect is particularly evident in the higher IoU gains of edge-sensitive categories shown in Table I, such as "cars" and "clutter." Furthermore, the explicit spatial alignment in FaPN prior to fusion mitigates artifacts introduced by naive upsampling and improves the geometric consistency of object boundaries. In addition, the ASFF module preserves small-target responses by adaptively selecting the most informative scale through pixel-wise weighting, rather than relying on fixed additive fusion. Collectively, these modules contribute to the overall improvement in segmentation performance of the proposed model.

2) Quantitative Comparison of Results on the LoveDA Dataset: As shown in Table II, the proposed model outperforms other comparison models across most metrics on the LoveDA dataset. The overall performance is superior, with the mIoU and aAcc for our method reaching 52.95% and 71.49%, respectively. These values surpass the best-performing CNN-based model, Deeplabv3+, by 1.96% in mIoU and 1.53% in aAcc. Additionally, our method outperforms the best Transformer-based model, Segformer, by 1.21% in mIoU and 1.09% in aAcc.

This further demonstrates that our approach, while effectively minimizing feature loss, better leverages the PVT to output multiscale features. In comparison to existing methods in rural and urban remote sensing image segmentation, particularly for roads and farmland, our approach significantly outperforms others. The method demonstrates excellent IoU performance, particularly when dealing with the notable differences in shape and appearance between these two categories. Compared with the average of other models in this experiment, the IoU of Water and Barren, which are categorized as small targets due to their fragmented or scattered distribution in this dataset, increased by an average of 4.7% and 3.7%, respectively. Meanwhile, the average IoU improvement for medium-scale targets including Road and Forest is 2.39% and 0.63%, respectively. For large-scale targets represented by the Background category, the improvement is 1.96%. Consequently, the overall effect of small-scale segmentation is more significant than other methods.

Similarly, when segmenting water and barren land, which are two categories in remote sensing images characterized by substantial intraclass variation and significant differences in spatial layout, shape, and size, our method achieves the best segmentation results. This highlights the method's robust adaptability in handling diverse and complex scene variations. Although the segmentation results for other categories did not achieve the highest IoU values, the differences are minimal. The redesigned encoder, which emphasizes edge segmentation, contributes to the optimal performance in mIoU and aAcc. Furthermore, the multiscale feature fusion decoder based on FaPN integrates target features from all output layers of the network, enhancing the details of target edges and improving segmentation accuracy. These results provide strong evidence of the excellent precision and generalization capabilities of the proposed method in remote sensing image segmentation.

C. Qualitative Comparison Diverse Methods

In order to comprehensively demonstrate the superior performance of the method proposed in this chapter in image segmentation tasks, this article provides a visual comparative analysis of segmentation results on the Potsdam dataset and Loveda dataset. The comparative analysis selected the two models PspNet and DeepLabv3+ with the best quantitative analysis performance based on CNN, as well as the two models SETR and Segformer with the best performance based on Transformer.

The comparison includes representative models based on CNN, namely PSPNet and DeepLabv3+, as well as transformer-based models like Ster and SegFormer. The segmentation outputs are displayed in Fig. 6. Dashed boxes mark regions where our method differs significantly from the other models, further highlighting its advantages.

From the comparative results, it is evident that PSPNet and DeepLabv3+ exhibit suboptimal performance in segmenting debris and trees, with coarse edge delineation and noticeable artifacts. Although SETR and SegFormer show improved overall segmentation quality, their accuracy in small target segmentation, such as vehicles, remains insufficient. SETR, in particular, struggles with blurry boundaries around vehicle contours. While SegFormer performs slightly better in vehicle segmentation, it exhibits substantial errors in building delineation, with some building regions mistakenly classified as impervious surfaces. In contrast, the proposed method demonstrates superior performance across multiple aspects. It achieves more precise boundary matching for debris and small targets like vehicles, significantly enhancing contour detail and completeness. Furthermore, in building segmentation, our approach exhibits stronger edge processing capabilities, producing smoother and more regular boundaries that better align with real-world structures.

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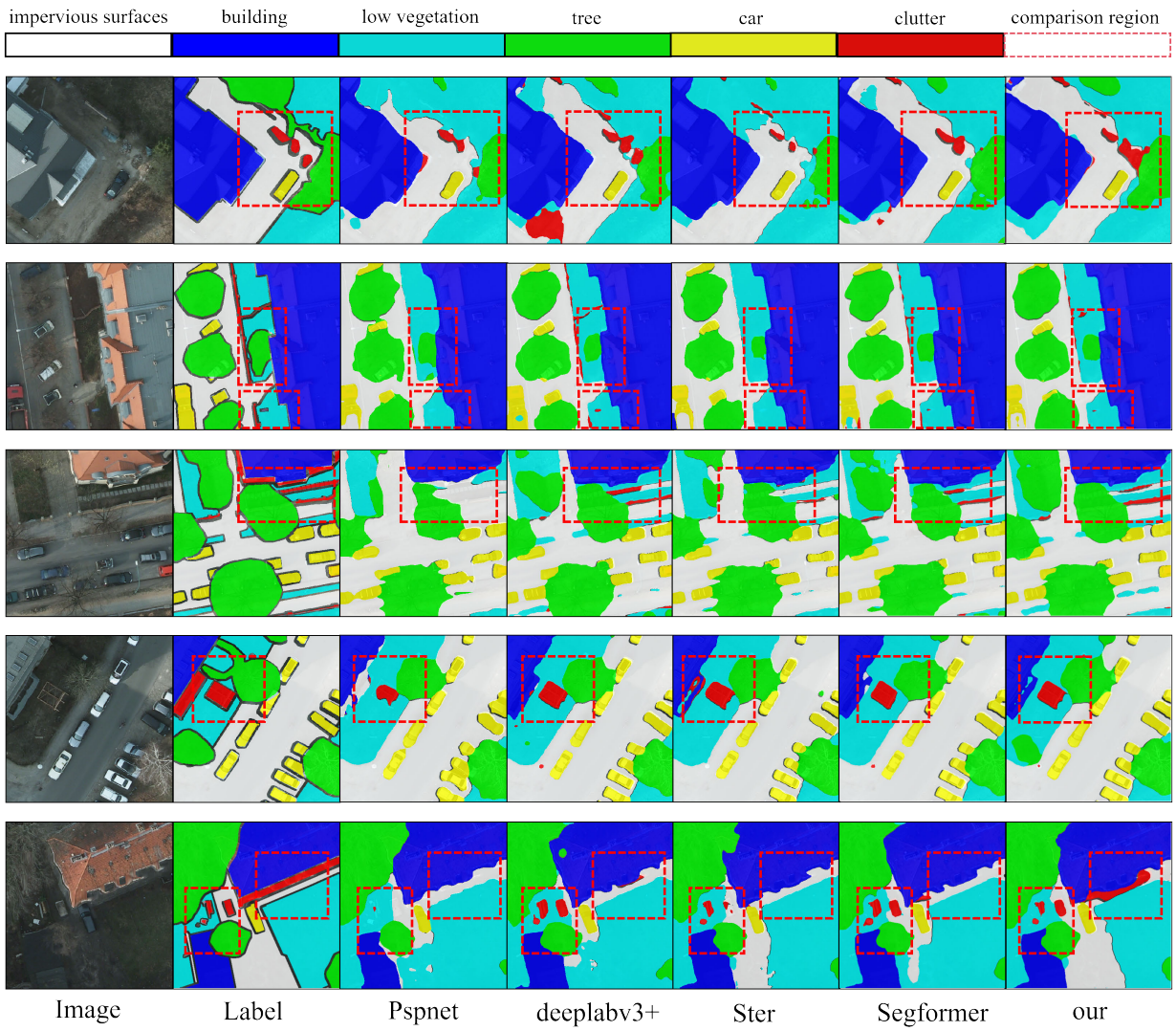


Fig. 6. Qualitative comparisons between our method and baseline on the Potsdam Dataset.

Our model not only surpasses others in overall segmentation accuracy but also excels in boundary detail preservation and small target segmentation. These marked improvements validate the effectiveness and robustness of the proposed method in enhancing multiclass segmentation accuracy and adapting to complex scenes.

In the visualizations of the LoveDA dataset, white indicates background, red denotes building, yellow represents road, blue denotes water, purple corresponds to barren, green indicates forest, and brown signifies agricultural area. Dashed boxes mark regions where our method differs significantly from the other models.

Fig. 7 presents a comparative analysis of segmentation results on the LoveDA dataset. Specifically, two well-performing CNN-based models (PspNet and DeepLabv3+) and two transformer-based models (SegFormer and Ster) are contrasted with our proposed model. Across this relatively large dataset, our model demonstrates more pronounced advantages.

In the comparison shown in the fourth row, the segmentation results of PspNet and DeepLabv3+ exhibit evident omissions of roads, whereas our method achieves complete and accurate segmentation of road regions. In the third-row comparison, both PspNet, DeepLabv3+, and the Stermodel fail to correctly predict the barren land category. In contrast, our approach successfully captures barren areas and provides more precise predictions for finer details.

This improvement can be attributed to the incorporation of the ASFF module, which optimizes boundary predictions during feature fusion, significantly enhancing the model's ability to capture image details and contours. Regarding building segmentation shown in the fifth row, our method produces sharper and clearer boundary delineations, while the competing models, SegFormer and Ster, exhibit blurred edges, with some instances even showing boundary merging.

Overall, the proposed method not only excels in overall segmentation accuracy but also significantly outperforms comparative models in fine-detail segmentation and boundary prediction, demonstrating its robustness and effectiveness in handling complex scenes.

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Fig. 7. Qualitative comparisons between our method and baseline on the LoveDA Dataset.

D. Ablation Study

To rigorously evaluate the contribution of each proposed component to semantic segmentation performance, we conducted a series of ablation experiments on two datasets. The experimental configuration and optimization parameters were kept consistent throughout, allowing us to isolate the impact of selectively removing specific modules. For the purpose of this analysis, EFM refers to the Edge-guided Perception Module, FFM denotes the Feature Fusion Module (incorporating FaPN), and ASFF stands for the ASFF module. Quantitative results, detailed in Tables III and IV, reveal that the EFM is indispensable for boundary representation, particularly for categories with irregular contours. As evidenced in Experiment 1, the removal of EFM precipitates a marked decline in performance for such classes; specifically, the IoU for Tree in the Potsdam dataset falls to 78.02%, while Forest in the LoveDA dataset drops to 38.01%. Furthermore, Building IoU decreases by 1.27%–2.19% across both datasets. This trend confirms that without explicit edge guidance, the network struggles to delineate sharp boundaries, resulting in indistinguishable edges. Notably, while the numerical drop for Clutter appears minor, visualization reveals a significant qualitative reduction in the recognizability of its contours.

Regarding cross-scale feature aggregation, the FFM proves critical for handling the extreme scale variations inherent in remote sensing imagery. In Experiment 2, where the FFM is excluded, the reliance on single-scale features fails to capture target diversity. Consequently, the IoU for Low Vegetation (typically large, amorphous regions) and Car (small targets) in the Potsdam dataset declines to 73.52% and 89.25%, respectively. A similar pattern emerges in the LoveDA dataset, where Agriculture and Barren suffer sharp drops to 55.48% and 27.34%, respectively, suggesting that FFM effectively prevents both the omission of small patches and semantic confusion within large, continuous areas. This fusion process is further refined by the ASFF module, which adaptively weights features. Its removal in Experiment 3 yields a specific impact; while Road (LoveDA) and Tree (Potsdam) see relatively mild decreases of 0.40%–0.52%, Agriculture in LoveDA drops significantly to 55.16%. This indicates that for categories with complex scale transitions, fixed fusion ratios are insufficient to balance detail preservation with semantic consistency.

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TABLE III
CATEGORY-SPECIFIC IOU VARIATIONS ON THE POTSDAM DATASET

expt.	EFM	FFM	ASFF	IoU per category (%)						mIoU(%)	aAcc(%)
				Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter		
1	×	✓	✓	87.55	92.15	76.55	78.02	91.33	46.00	78.60	90.18
2	✓	×	✓	86.03	91.53	73.52	75.89	89.25	42.66	76.48	89.88
3	✓	✓	×	87.79	93.21	77.15	79.54	92.28	43.73	78.95	90.46
4	✓	×	×	84.02	89.02	70.09	72.59	85.82	46.18	74.62	85.10
5	×	×	✓	85.52	90.52	71.09	73.59	87.32	44.24	75.38	86.72
6	✓	✓	✓	**87.97**	**93.42**	**77.35**	**79.94**	**92.48**	**46.70**	**79.64**	**91.39**

TABLE IV
CATEGORY-SPECIFIC IOU VARIATIONS ON THE LOVEDA DATASET

expt.	EFM	FFM	ASFF	IoU per category (%)							mIoU(%)	aAcc(%)
				Background	Building	Road	Water	Barren	Forest	Agriculture		
1	×	✓	✓	52.83	56.73	53.27	70.12	30.04	38.01	57.47	51.21	70.44
2	✓	×	✓	51.33	55.03	50.93	68.32	27.34	36.81	55.48	49.32	68.66
3	✓	✓	×	53.33	58.33	54.42	70.62	31.24	39.71	55.16	51.83	70.81
4	✓	×	×	53.03	52.33	48.53	66.22	24.84	33.61	52.96	47.36	64.26
5	×	×	✓	52.33	53.43	49.63	67.12	25.94	34.71	54.73	48.27	65.09
6	✓	✓	✓	**53.83**	**58.92**	**54.94**	**71.22**	**31.84**	**40.51**	**59.38**	**52.95**	**71.49**

TABLE V
COMPARISON WITH OTHER COMPARATIVE NETWORKS

Backbone	Method	mIoU(%)	Parameters(M)	Complexity(G)	FPS
CNN-base	FCN [5]	49.28	47.13	1581.58	5.1
	DeepLab V3+ [7]	50.99	65.74	1609.98	4.9
	Pspnet [63]	50.29	46.6	2048.73	3.9
	DDRnet [64]	49.89	57.13	1725.12	4.6
Transformer-base	Ster [19]	49.79	309.1	1893.64	4.2
	Segmenter [65]	50.48	102.37	1895.36	4.2
	Segformer [66]	51.74	27.25	1795.67	4.4
	SAN [67]	50.2	8.4	1664.51	4.8
	Ours	52.95	42.13	1768.41	4.6

The synergy between these modules is fundamental for robust segmentation, as demonstrated when multiple components are removed. The simultaneous removal of FFM and ASFF (Experiment 4) causes a severe performance drop due to the lack of cross-scale alignment; notably, Tree and Low Vegetation in Potsdam decline by over 7%, while Barren in LoveDA falls by 7.00%. The degradation is even more pronounced in Experiment 5, where both EFM and FFM are removed. The combined loss of edge guidance and multi-scale fusion leads to substantial deterioration, with low vegetation (Potsdam) and building (LoveDA) decreasing by over 6% and 5%, respectively, and the overall mIoU dropping by 4.26%–4.68%. These findings confirm that the "Edge-Guided" and "Multi-Scale" components are mutually reinforcing rather than redundant, ensuring both boundary fidelity and semantic consistency.

These quantitative findings are further corroborated by the visual comparisons presented in Fig. 8. Experiment 1, which integrates only FFM and ASFF, suffers from the absence of edge guidance, leading to evident irregular boundary delineation around buildings and trees in the second and third rows. Experiment 2, lacking the FFM, exhibits noticeable object segmentation errors and semantic confusion due to insufficient cross-scale context. Similarly, Experiment 3, without ASFF, fails to adaptively weight features, resulting in the loss of key target details and unclear separation between adjacent classes like buildings and vehicles. Performance degradation reaches its peak in single-module configurations: Experiment 4 (EFM only) struggles with semantic consistency, leading to unclear recognition of trees and clutter in the first row, while Experiment 5 (ASFF only) shows significant misclassification, erroneously identifying buildings and background regions as clutter. In contrast, Experiment 6, which represents our proposed method, effectively integrates all three modules. The resulting visualizations demonstrate greatly enhanced detail in target edges and small objects, effectively mitigating the adverse effects of shadows and background blurring, thereby validating the indispensability and synergistic effectiveness of the proposed architecture.

E. Efficiency Evaluation

To further explore the complexity and parameter efficiency of the proposed method in remote sensing image semantic segmentation, and to validate its effectiveness, we conducted a

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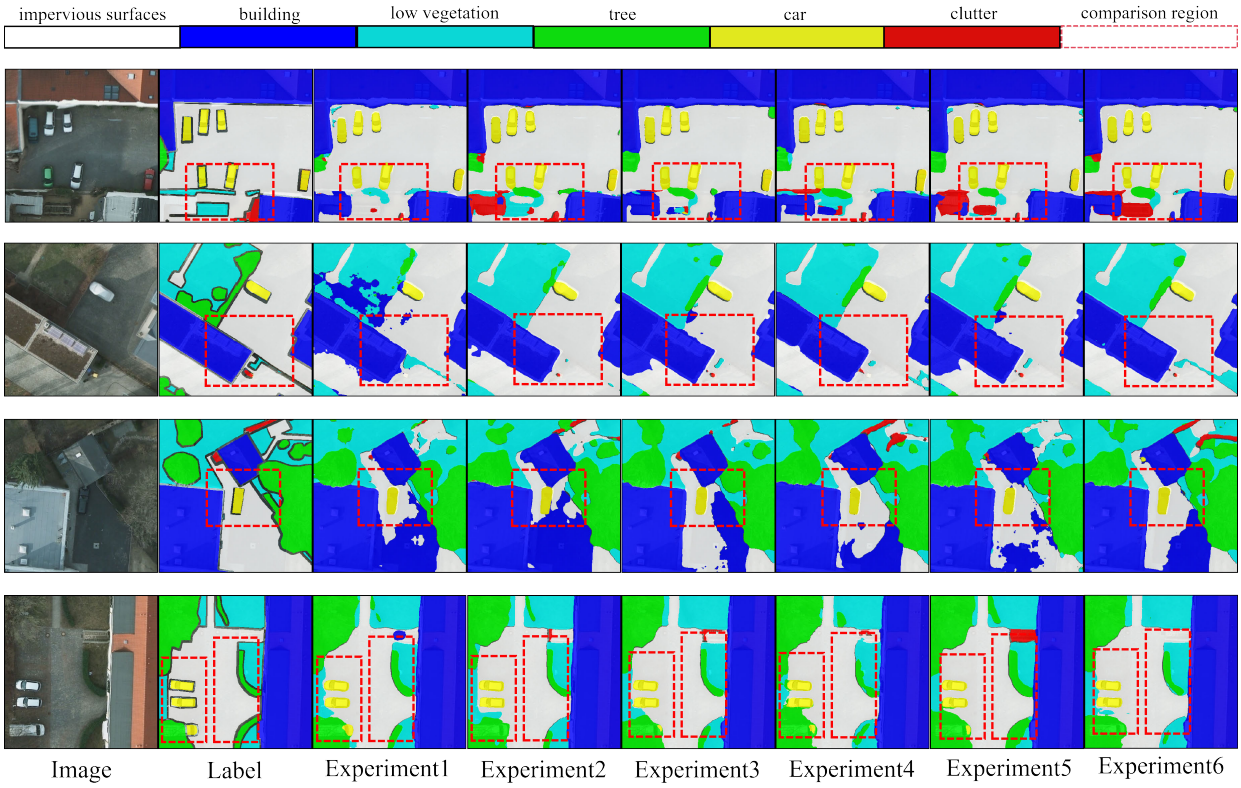


Fig. 8. Visualization results of ablation experiments on the Potsdam dataset.

model complexity analysis on the LoveDA dataset. The experimental setup and training optimization parameters remained consistent with the comparison experiments. The detailed results are shown in Table V.

Our proposed model achieved the highest mIoU of 52.95%, significantly outperforming other comparative models. Specifically, compared to DeepLabV3+(50.99%), Pspnet(50.29%), SegFormer(51.74%), and SAN(50.20%), our model improved by 1.96%, 2.66%, 1.21%, and 2.75%, respectively. This notable enhancement demonstrates the superior feature extraction and representation capabilities of our approach.

In terms of parameter count, our model contains 42.13 million parameters, which is substantially lower than Transformer-based models such as Ster (309.1M) and Segmenter (102.37M), and more efficient than traditional models like DeepLabV3+ (65.74M) and Pspnet (46.6M). Although SAN has the fewest parameters (8.40M), its mIoU remains at 51.74%, which is lower than that of our model. This indicates that our model achieves higher precision while maintaining a relatively low parameter count, which is a critical factor for practical deployment.

Regarding computational complexity, our model requires 1768.41 GFLOPs. While not the lowest among the compared models, it is more efficient than Pspnet (2048.73 GFLOPs) and some Transformer-based models, such as SETR (1893.64 GFLOPs). Moreover, when considering inference speed, our model achieves 4.6 FPS, which is higher than Pspnet (3.9 FPS) and comparable to SAN (4.8 FPS). CNN-based methods such as FCN reach slightly higher FPS (5.1) but suffer from much lower accuracy (49.28%), showing an unfavorable tradeoff. Coupled with its top performance in mIoU, our model demonstrates an excellent balance among accuracy, computational efficiency, and inference speed. These advantages underscore its practical value, effectively balancing high segmentation performance with resource consumption and making it highly suitable for real-world applications.

V. Conclusion

In summary, this study presents an engineering-integrated framework tailored for high-resolution remote sensing segmentation. By anchoring the architecture on a PVT backbone augmented by the EFM, we explicitly target the restoration of boundary details often compromised during downsampling. Complementing this, the decoder—leveraging a FaPN with ASFF—ensures that these recovered details are rigorously aligned with high-level semantics to resolve scale discrepancies. Quantitative benchmarks on the Potsdam and LoveDA datasets confirm that this synergistic design effectively reconciles the tradeoff between scale variations and boundary precision, yielding robust performance improvements over state-of-the-art baselines.

However, limitations persist in scenarios involving severe occlusion or strong shadows, where boundary cues alone are insufficient to recover full object extents. Furthermore, distinguishing structurally similar categories remains challenging at the current resolution. Future iterations will address these bottlenecks by incorporating auxiliary boundary supervision and exploiting multispectral data to enhance class separability in complex environments.

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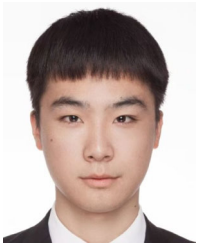
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